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Authentication Protocols

Many different authentication protocols are used in networking to verify the identity of devices and users. Those protocols are essential because they provide a secure and standardized way of **verifying the identity** of users, devices, and other entities in a network. Without authentication protocols, it would be difficult to securely and reliably identify entities in a network, making it easy for attackers to gain unauthorized access and potentially compromise the network.

Authentication protocols also provide a means for **securely exchanging information** between entities in a network which we will discuss briefly. This is important for ensuring the confidentiality and integrity of sensitive data and preventing unauthorized access and other security threats. Let us take a look at a few most commonly used authentication protocols:

Protocol	Description
Kerberos	Key Distribution Center (KDC) based authentication protocol that uses tickets in domain environments.

Protocol	Description
SRP	This is a password-based authentication protocol that uses cryptography to protect against eavesdropping and man-in-the-middle attacks.
SSL	A cryptographic protocol used for secure communication over a computer network.
TLS	TLS is a cryptographic protocol that provides communication security over the internet. It is the successor to SSL.
OAuth	An open standard for authorization that allows users to grant third-party access to their web resources without sharing their passwords.
OpenID	OpenID is a decentralized authentication protocol that allows users to use a single identity to sign in to multiple websites.
SAML	Security Assertion Markup Language is an XML-based standard for securely exchanging authentication and authorization data between parties.
2FA	An authentication method that uses a combination of two different factors to verify a user's identity.
FIDO	The Fast IDentity Online Alliance is a consortium of companies working to develop open standards for strong authentication.
PKI	PKI is a system for securely exchanging information based on the use of public and private keys for encryption and digital signatures.

Protocol	Description
SSO	An authentication method that allows a user to use a single set of credentials to access multiple applications.
MFA	MFA is an authentication method that uses multiple factors, such as something the user knows (a password), something the user has (a phone), or something the user is (biometric data), to verify their identity.
PAP	A simple authentication protocol that sends a user's password in clear text over the network.
CHAP	An authentication protocol that uses a three-way handshake to verify a user's identity.
EAP	A framework for supporting multiple authentication methods, allowing for the use of various technologies to verify a user's identity.
SSH	This is a network protocol for secure communication between a client and a server. We can use it for remote command-line access and remote command execution, as well as for secure file transfer. SSH uses encryption to protect against eavesdropping and other attacks and can also be used for authentication.
HTTPS	This is a secure version of the HTTP protocol used for communication on the internet. HTTPS uses SSL/TLS to encrypt communication and provide authentication, ensuring that third parties cannot intercept and read the transmitted data. It is widely used for secure communication over the internet, particularly for web browsing.

Protocol	Description
LEAP	LEAP is a wireless authentication protocol developed by Cisco. It uses EAP to provide mutual authentication between a wireless client and a server and uses the RC4 encryption algorithm to encrypt communication between the two. Unfortunately, LEAP is vulnerable to dictionary attacks and other security vulnerabilities and has been largely replaced by more secure protocols such as EAP-TLS and PEAP.
PEAP	PEAP on the other hand is a secure tunneling protocol used for wireless and wired networks. It is based on EAP and uses TLS to encrypt communication between a client and a server. PEAP uses a server-side certificate to authenticate the server and can also be used to authenticate the client using various methods, such as passwords, certificates, or biometric data. PEAP is widely used in enterprise networks for secure authentication.

For example, LEAP and PEAP may be used to authenticate wireless clients when they access the wireless network or to authenticate remote employees connecting to the network via a VPN. In these cases, using SSH or HTTPS would be challenging to implement, as they are designed for different purposes. In terms of security, **PEAP** is generally more secure than LEAP because it uses a server-side public key certificate to authenticate the server and encrypting **MSCHAPv2** hash, while **LEAP** relies on a shared secret that is negotiated between the client and the server and does **not** encrypt **MSCHAPv2** hashes. PEAP also supports the use of stronger encryption algorithms, such as AES and 3DES, for encrypting the authentication information, whereas LEAP uses the weaker RC4 algorithm.

However, both protocols are vulnerable to specific attacks and have been largely replaced by more secure protocols such as EAP-TLS.

As physical connections, protocols with **SSL** or **TLS** are used by default, such as **SSH** or **HTTPS**. Both protocols use robust encryption algorithms to protect the authentication information transmitted between the client and the server. This helps to prevent interception or to tamper the authentication data, which is essential for maintaining the security of the authentication process. Also, they support using digital certificates and **PKI** for authenticating the server to the client. This ensures that the client can verify the server's identity and helps to prevent MITM attacks. SSH and HTTPS are widely used and well-supported protocols, with implementations available for various operating systems and devices and makes them easy to use and deploy in a variety of environments.



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